

Logistic (Logit) Regression

CJ 702: Advanced Criminal Justice Statistics

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1 Setting up your environment

```
# Load Packages  
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2     3.5.1      v tibble     3.2.1  
## v lubridate   1.9.4      v tidyr      1.3.1  
## v purrr       1.0.2  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()    masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

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```
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##   recode
##
## The following object is masked from 'package:purrr':
##
##   some
```

```
library(ggpubr)
library(Hmisc)
```

```
##
## Attaching package: 'Hmisc'
##
## The following objects are masked from 'package:dplyr':
##
##   src, summarize
##
## The following objects are masked from 'package:base':
##
##   format.pval, units
```

```
# ===== #
```

```
# Load the NCVS dataset we have been working with:
```

```
person <- readRDS("../Data/person.rds")
```

```
# Check your data:
```

```
head(person)
```

```
## # A tibble: 6 x 19
##   YEAR YEARQ IDPER      IDHH   AGE  SEX  V3020 WGTPER  YIH VIC_LOC WGTVIC_V
##   <dbl> <fct> <fct>      <fct> <fct> <fct> <fct>  <dbl> <dbl> <chr>      <dbl>
## 1  2000  001   2000966984 200099~ 40-49 Male  Coll~  1063.    10 <NA>        NA
## 2  2000  001   2000951294 200099~ 40-49 Fema~  Coll~   894.     9 <NA>        NA
## 3  2000  001   2000470356 200099~ 12-17 Male  Elem~  1317.     9 <NA>        NA
## 4  2000  001   2000205990 200016~ 35-39 Male  Coll~  1093.     4 <NA>        NA
## 5  2000  001   2000361146 200016~ 30-34 Fema~  Coll~  1101.     4 Home       2202.
## 6  2000  001   2000879996 200073~ 40-49 Male  High~  1063.     6 <NA>        NA
## # i 8 more variables: VIOLENT <dbl>, WGTVIC_NV <dbl>, NONVIOLENT <dbl>,
## #   ADJINC_WT_V <dbl>, VLNT_WGT <dbl>, ADJINC_WT_NV <dbl>, NVLNT_WGT <dbl>,
## #   EDUC <fct>
```

```
# Take note of the variables:
```

```
## ID: Person ID (numeric)
```

```
## IDHH: Household ID (numeric)
## PER_WGT: Person Weight (numeric)
## VIOLENT: Violent victimization count (numeric, count, ratio)
## VLNT_WGT: Violent victimization weight (numeric)
## NONVIOLENT: Nonviolent victimization count (numeric, count, ratio)
## NVLNT_WGT: Nonviolent victimization weight (numeric)
# YIH: Years in household (numeric, years, interval)
# EDUC: Education level (factor, ordinal)
# AGE: Age (factor, years, ordinal)
# SEX: Sex (factor, nominal)

# Check for missingness:
missing <- person %>%
  filter(!complete.cases(VIOLENT, NONVIOLENT,
                          YIH, EDUC, AGE, SEX)) %>%
  nrow()
n <- person %>% nrow()
missing / n
```

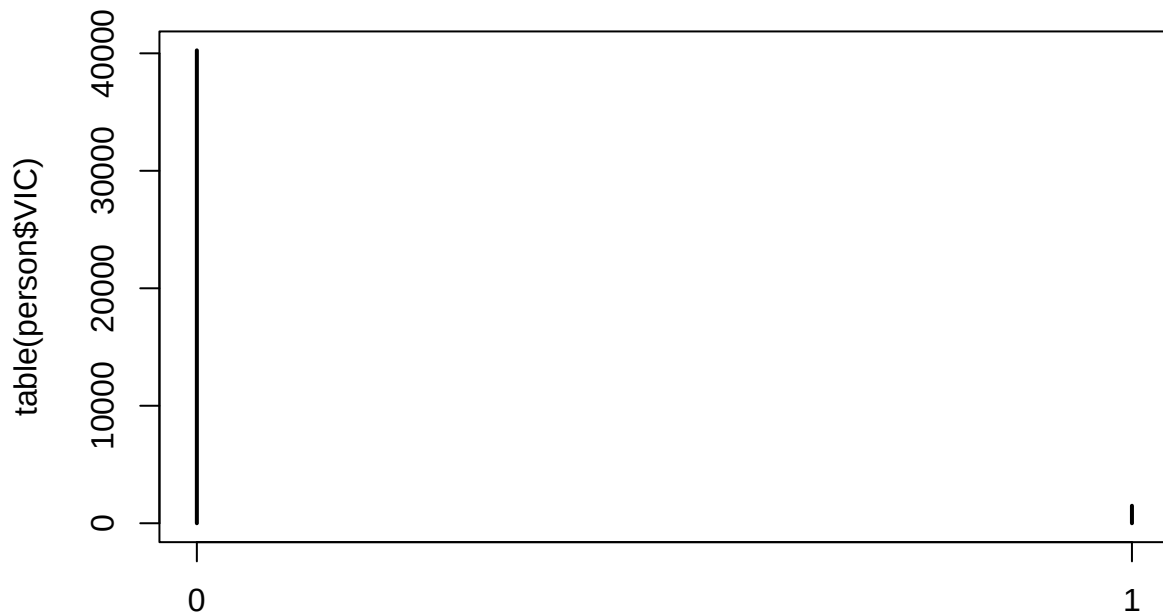
```
## [1] 0.08812478
```

```
# Satisfied with sufficiently low missingness,
# you can perform listwise deletion:
person <- person %>%
  filter(complete.cases(VIOLENT, NONVIOLENT,
                        YIH, EDUC, AGE, SEX))

# Create your binary dependent variable (victimization)
person <- person %>%
  mutate(VIC = as.numeric((VIOLENT > 0) | (NONVIOLENT > 0)))
```

2 Descriptives and visualizing binary variables

```
# You can generate a simple plot of the Bernoulli distribution of your
# dependent variable with the plot() and table() functions:
person$VIC |> table() |> plot()
```



```
# This is useful for your own diagnostics, and understanding what proportion
# of respondents in your data were victimized. However, it's not analytically
# interesting and best described with the mean() function.
```

```
# Remember, the mean() of a Bernoulli random variable is the proportion, 'p',
# of observations with the affirmative / TRUE / "1" response:
person$VIC |> mean()
```

```
## [1] 0.03567151
```

```
# You can find the variance, which is defined as p * (1 - p),
# with the var() function. However, like the plot, you typically
# wouldn't include this in a publication (it doesn't tell you much!)
person$VIC |> var()
```

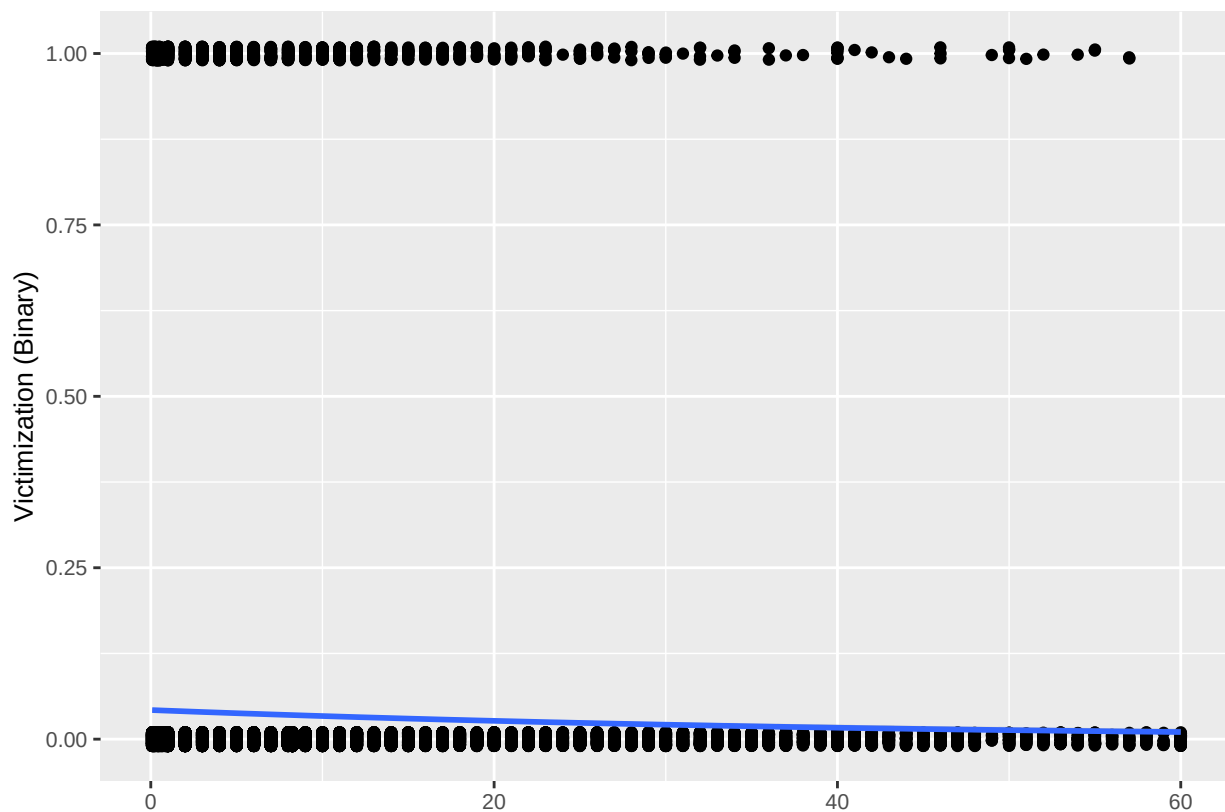
```
## [1] 0.03439987
```

```
# The table() function by itself will provide you the frequencies for
# the variable:
person$VIC |> table()
```

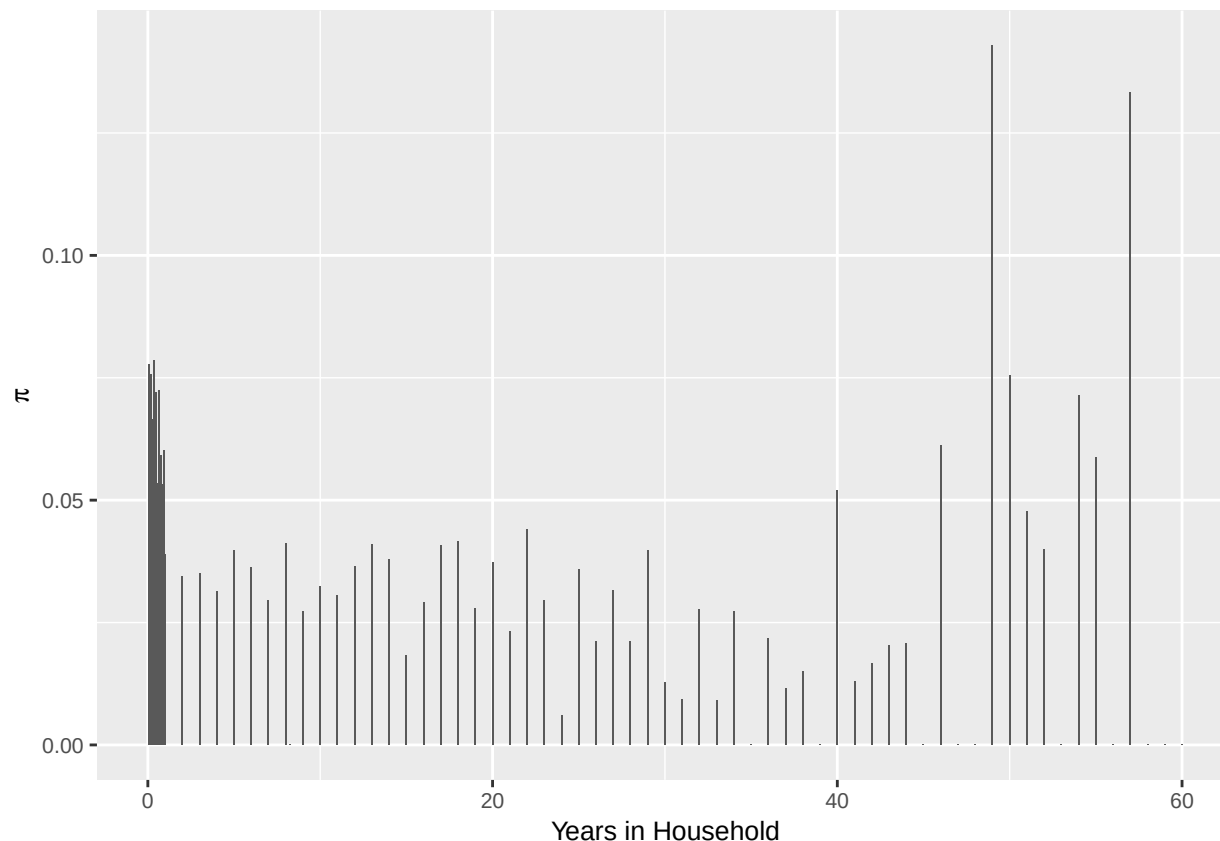
```
##
##      0      1
## 40253  1489
```

```
# Bivariate Graphs
## Visualizing the relationship between two variables as a scatter plot:
ggplot(person, aes(x = YIH, y = VIC)) +
  geom_jitter(width = 0.01, height = 0.01) +
  geom_smooth(method = "glm",
              method.args = list(family = "binomial"),
              se = TRUE) +
  labs(x = " ",
       y = "Victimization (Binary)") +
  theme(text = element_text(size = 10))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Visualizing the relationship between two variables as a bar chart:
person %>%
  select(YIH, VIC) %>%
  group_by(YIH) %>%
  summarise(`pi` = mean(VIC)) %>%
  ggplot(aes(x = YIH, y = `pi`)) +
  geom_bar(stat = "identity") +
  labs(x = "Years in Household",
       y = expression(pi)) +
  theme(text = element_text(size = 10))
```



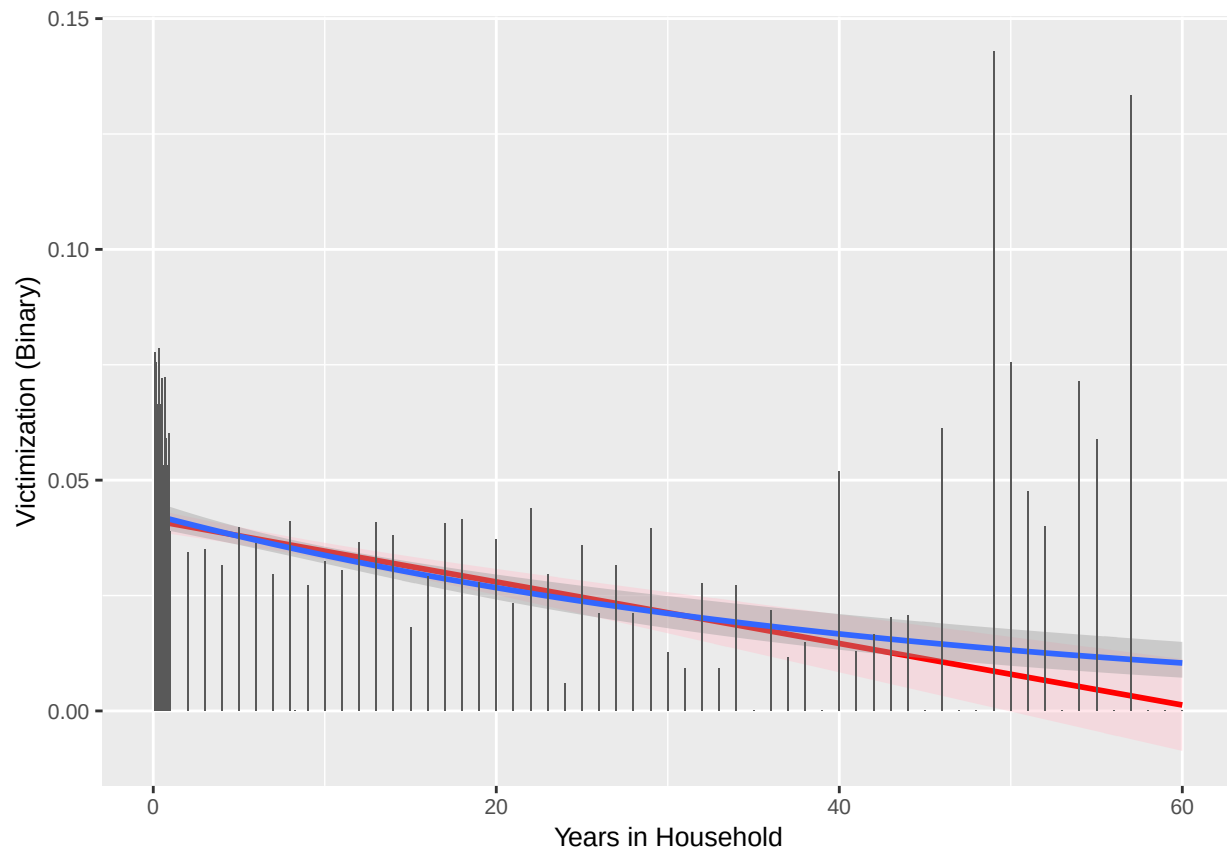
```
## Combining both approaches:
### Prepare the bar data using what we know about  $\pi = f(x)$ :
```

```
person <- person %>%
  group_by(YIH) %>%
  mutate(n = n(),
         `pi` = mean(VIC),
         `pi/n` = `pi` / n) %>%
  ungroup()
```

```
### Build the combined plot:
```

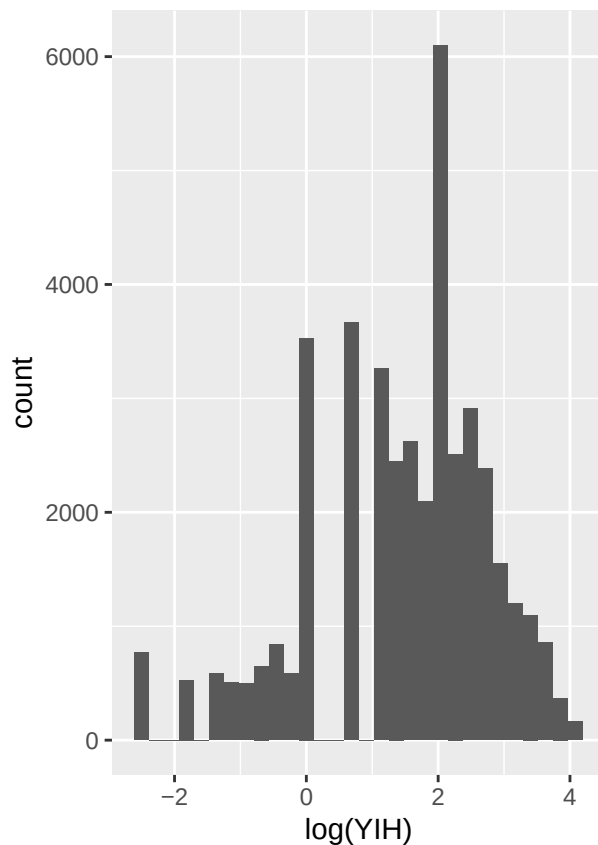
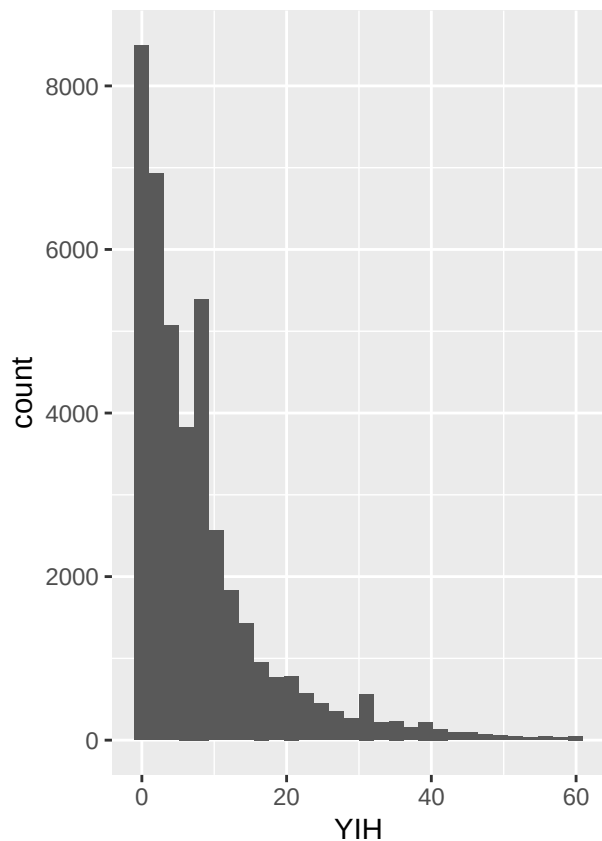
```
ggplot(person, aes(x = YIH, y = VIC)) +
  geom_smooth(method = "lm",
             se = TRUE,
             color = "red",
             fill = "pink") +
  geom_smooth(method = "glm",
             method.args = list(family = "binomial"),
             se = TRUE) +
  geom_bar(aes(y = `pi/n`), stat = "identity") +
  labs(x = "Years in Household",
       y = "Victimization (Binary)") +
  theme(text = element_text(size = 10))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Looking at the previous plots, you may notice that the
## "years in household" variable is right-skewed.
## To 'normalize' the variable, we can log-transform it:
ggarrange(ggplot(person, aes(x = YIH)) + geom_histogram(),
          ggplot(person, aes(x = log(YIH))) + geom_histogram())
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

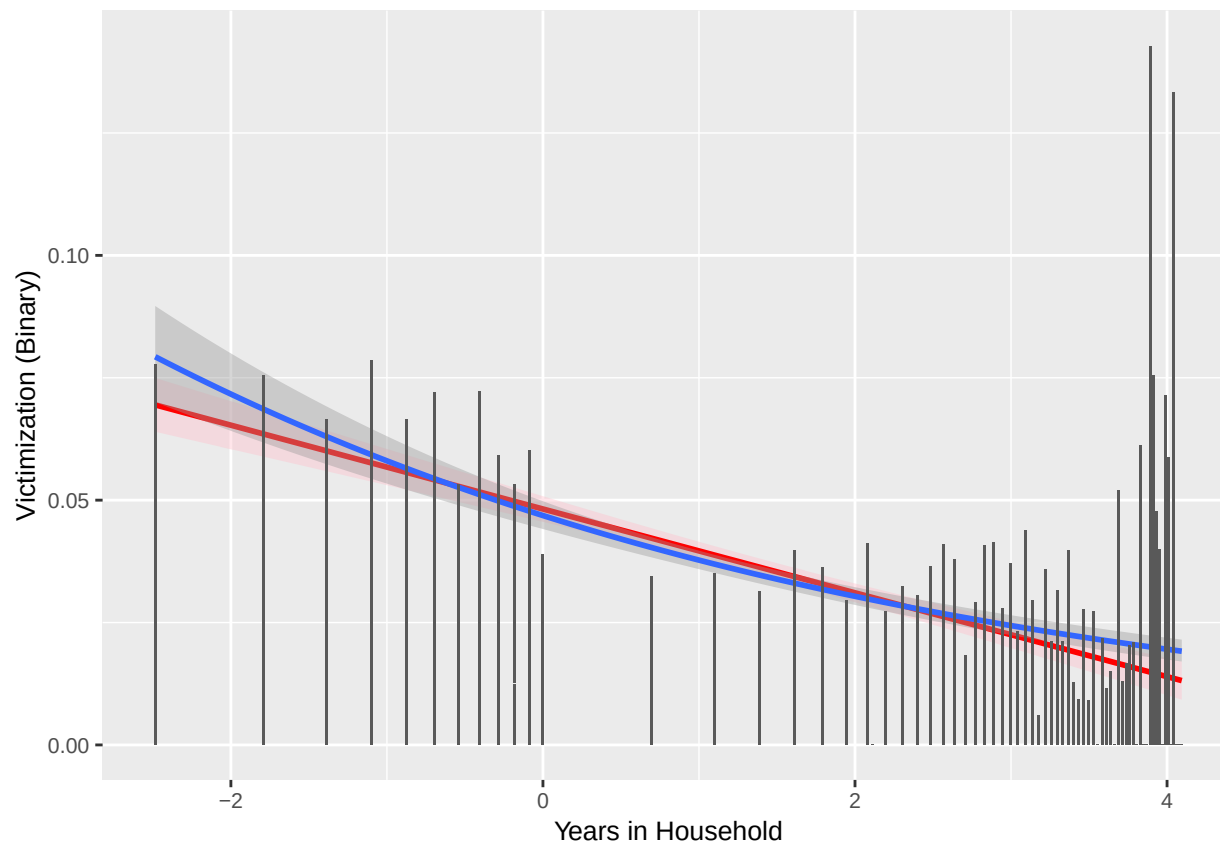


```
## We can apply this log transformation to the code for the
## above combined plot to examine how it affects bivariate
## model fit:
```

```
person <- person %>%
  group_by(log(YIH)) %>%
  mutate(n = n(),
         `pi` = mean(VIC),
         `pi/n` = `pi` / n) %>%
  ungroup()

ggplot(person, aes(x = log(YIH), y = VIC)) +
  geom_smooth(method = "lm",
             se = TRUE,
             color = "red",
             fill = "pink") +
  geom_smooth(method = "glm",
             method.args = list(family = "binomial"),
             se = TRUE) +
  geom_bar(aes(y = `pi/n`), stat = "identity") +
  labs(x = "Years in Household",
       y = "Victimization (Binary)") +
  theme(text = element_text(size = 10))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
```



3 Estimating binomial generalized linear models (logit)

```
# Let's start by fitting a linear probability model using the lm() function:
summary(m1 <- lm(VIC ~ log(YIH) +                                # Log Years in Household
                I(log(YIH)^2) +                                # Log Years in Household^2
                scale(as.numeric(AGE)) +                       # Age (Ordinal)
                scale(as.numeric(EDUC)) +                       # Education (Ordinal)
                SEX,                                           # Sex (Binary)
                data = person))
```

```
##
## Call:
## lm(formula = VIC ~ log(YIH) + I(log(YIH)^2) + scale(as.numeric(AGE)) +
##     scale(as.numeric(EDUC)) + SEX, data = person)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.09932 -0.03856 -0.03157 -0.02626  0.98197
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.0422185   0.0017513  24.107  < 2e-16 ***
## log(YIH)     -0.0119817   0.0009929 -12.068  < 2e-16 ***
```

```
## I(log(YIH)^2)          0.0023538  0.0003871   6.081 1.20e-09 ***
## scale(as.numeric(AGE)) -0.0048463  0.0010320  -4.696 2.66e-06 ***
## scale(as.numeric(EDUC)) 0.0014889  0.0009607   1.550  0.1212
## SExFemale              0.0031691  0.0018149   1.746  0.0808 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.185 on 41736 degrees of freedom
## Multiple R-squared:  0.00511,    Adjusted R-squared:  0.004991
## F-statistic: 42.88 on 5 and 41736 DF,  p-value: < 2.2e-16

## Model Specification:
### Pr(VIC = 1) ~ b0 + b1(YIH) + b2(YIH^2) + b3(AGE) + b4(EDUC) + b5(SEX) + e

## Interpretation (same as OLS, but the DV is a probability):
### b0: the intercept, the average value of the DV when all IVs are 0.
### bk: the average change in the probability of the DV (1),
###      for each interval increase in the IV, controlling for the other IVs.
### Pr(>|t|): P-value, probability of observing the current (or a more extreme)
###           effect size under the assumption that the null hypothesis is true.
### Degrees of freedom: n - (k + 1); n = # obs; k = # IVs.
### R-squared: Proportion of variance in the DV explained by the IVs.
### F-test: Overall model significance.

## For all of the reasons discussed in class, and demonstrated in the figures
## you generated in the previous section of this R module, it is typically
## ill-advised to fit a linear probability model.

# ===== #

# Now, let's fit a logit model with the same specification, but
# Pr(VIC = 1) becomes log(VIC / (1 - VIC)):

summary(m2 <- glm(VIC ~ log(YIH) + I(log(YIH)^2) +
  scale(as.numeric(AGE)) +
  scale(as.numeric(EDUC)) +
  SEX,
  data = person,
  family = binomial(logit)))
# Log Years in Household
# Log Years in Household^2
# Age (Ordinal)
# Education (Ordinal)
# Sex (Binary)

##
## Call:
## glm(formula = VIC ~ log(YIH) + I(log(YIH)^2) + scale(as.numeric(AGE)) +
##      scale(as.numeric(EDUC)) + SEX, family = binomial(logit),
##      data = person)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.17674    0.05053 -62.865 < 2e-16 ***
## log(YIH)       -0.24488    0.02133 -11.481 < 2e-16 ***
## I(log(YIH)^2)    0.03492    0.01007   3.466 0.000528 ***
## scale(as.numeric(AGE)) -0.13625    0.03003  -4.537 5.7e-06 ***
## scale(as.numeric(EDUC))  0.04197    0.02822   1.488 0.136857
```

```
## SEXFemale          0.09214    0.05318    1.733 0.083134 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 12851  on 41741  degrees of freedom
## Residual deviance: 12670  on 41736  degrees of freedom
## AIC: 12682
##
## Number of Fisher Scoring iterations: 6
```

```
## Note that the only changes to the code are: (1) the function, which
## changes from lm() [linear model] to glm() [generalized linear model],
## and we introduce the "family =" option with the "binomial(logit)"
## link function.
```

```
# Interpreting the results
## Log Odds
summary(m2)
```

```
##
## Call:
## glm(formula = VIC ~ log(YIH) + I(log(YIH)^2) + scale(as.numeric(AGE)) +
##      scale(as.numeric(EDUC)) + SEX, family = binomial(logit),
##      data = person)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -3.17674    0.05053  -62.865 < 2e-16 ***
## log(YIH)        -0.24488    0.02133  -11.481 < 2e-16 ***
## I(log(YIH)^2)    0.03492    0.01007    3.466 0.000528 ***
## scale(as.numeric(AGE)) -0.13625    0.03003   -4.537 5.7e-06 ***
## scale(as.numeric(EDUC)) 0.04197    0.02822    1.488 0.136857
## SEXFemale        0.09214    0.05318    1.733 0.083134 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 12851  on 41741  degrees of freedom
## Residual deviance: 12670  on 41736  degrees of freedom
## AIC: 12682
##
## Number of Fisher Scoring iterations: 6
```

```
### Intercept: -3.18      When all IVs are 0, we expect the average
###                      log odds of victimization to be -3.18.
```

```
### AGE: -0.14           For each standard deviation increase in age, we expect
###                      an average reduction of 0.14 in the log odds
###                      of victimization, net of control variables.
```

```
### SEX (Female): 0.09    On average, women are expected to score 0.09
###                      greater than men on the log-odds of victimization.
```

```
### Note that these interpretations are all a little clunky. This is because
### there is no real 'meaningful' interpretation for the log-odds.
### It is an unintuitive transformation.
```

```
## Odds Ratios
```

```
### Conveniently, you can use this handy line of code to simultaneously
### exponentiate your coefficients AND confidence intervals!
exp(cbind(coef(m2), confint(m2)))
```

```
## Waiting for profiling to be done...
```

```
##                      2.5 %    97.5 %
## (Intercept)          0.04172157 0.03775226 0.04602326
## log(YIH)             0.78279586 0.75118786 0.81672662
## I(log(YIH)^2)        1.03553652 1.01509692 1.05599809
## scale(as.numeric(AGE)) 0.87262350 0.82273087 0.92551763
## scale(as.numeric(EDUC)) 1.04286828 0.98671379 1.10211919
## SEXFemale            1.09652154 0.98812791 1.21719159
```

```
### Intercept: 0.04    When all IVs are 0, we expect the average
###                   odds of victimization to be 0.04.
```

```
### AGE: 0.87          For each standard deviation increase in age, we expect
###                   a 0.87 factor change in victimization likelihood/
###                   a 13 percent reduction in the odds of victimization.
```

```
### SEX (Female): 1.10    On average, women are expected to report at
###                   least one victimization 10% more frequently than men.
```

4 Post-estimation functions and visualization

```
# Predicted Probabilities
```

```
## Whole sample:
```

```
head(predict(m2, type = "response"))
```

```
##          1          2          3          4          5          6
## 0.02703490 0.02983363 0.03291636 0.03176214 0.03677757 0.02702775
```

```
## Typical / interesting individuals:
```

```
pred_prob <- function(y){
  exp(y) / (1 + exp(y))
}
```

```
### Keep in mind the order of your variables / coefficients:
```

```
#### 1. Intercept
```

```
#### 2. Years in Household
#### 3. Years in Household (Squared)
#### 4. Age (Centered, Z-Score)
#### 5. Education (Centered, Z-Score)
#### 6. Sex (Female = 1, Binary)
```

```
### Men (0 years in home, average age and education level):
sum(coef(m2) * c(1, 0, 0, 0, 0, 0)) %>%
  pred_prob()
```

```
## [1] 0.04005059
```

```
### Women (0 years in home, average age and education level):
sum(coef(m2) * c(1, 0, 0, 0, 0, 1)) %>%
  pred_prob()
```

```
## [1] 0.04374722
```

```
### Women w. a PhD (0 years in home, average age):
sum(coef(m2) * c(1,
  0,
  0,
  0,
  person$EDUC %>% as.numeric() %>% scale() %>% max(),
  1)) %>%
  pred_prob()
```

```
## [1] 0.04948469
```

```
## Testing the effect of specific parameters on the sample:
### The following code will give you the predicted probabilities
### for the whole sample (maintaining their observed scores for
### most variables), but treat all observations as FEMALE.
### This is achieved by "forcing" the "SEX" variable to be "Female".
### If you View() the ppf object, you can verify that all observations
### are treated as "Female" for the purpose of generating predictions.
### You can do this for any regression, and any variable!
```

```
ppf <- data.frame("YIH" = person$YIH,
  "AGE" = person$AGE,
  "EDUC" = person$EDUC,
  "SEX" = as.factor("Female"))
head(fm_pp <- predict(m2, newdata = ppf, type = "response"))
```

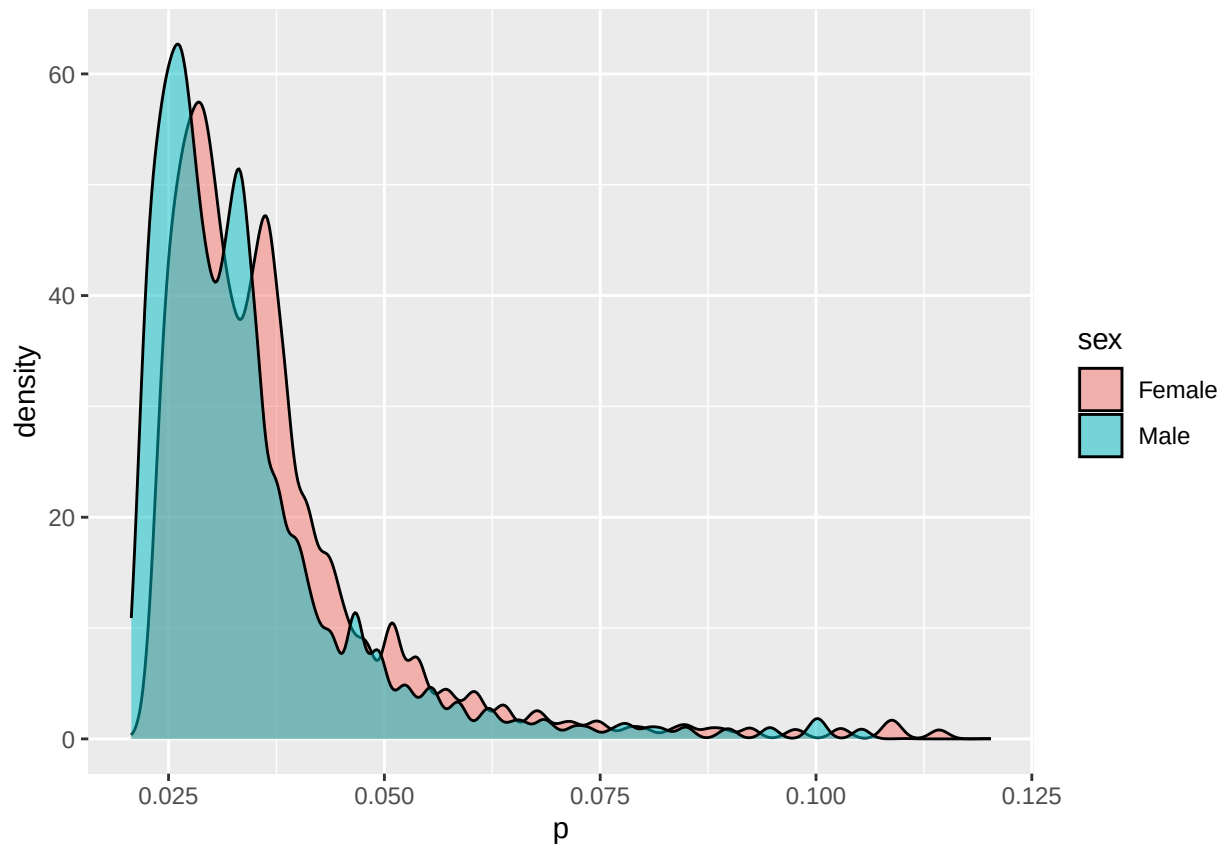
```
##           1           2           3           4           5           6
## 0.02956720 0.02983363 0.03597919 0.03472142 0.03677757 0.02955940
```

```
### If we generate the same for men, we could look at how the distribution
### changes when comparing men and women:
```

```
ppm <- data.frame("YIH" = person$YIH,
  "AGE" = person$AGE,
  "EDUC" = person$EDUC,
  "SEX" = as.factor("Male"))
head(m_pp <- predict(m2, newdata = ppm, type = "response"))
```

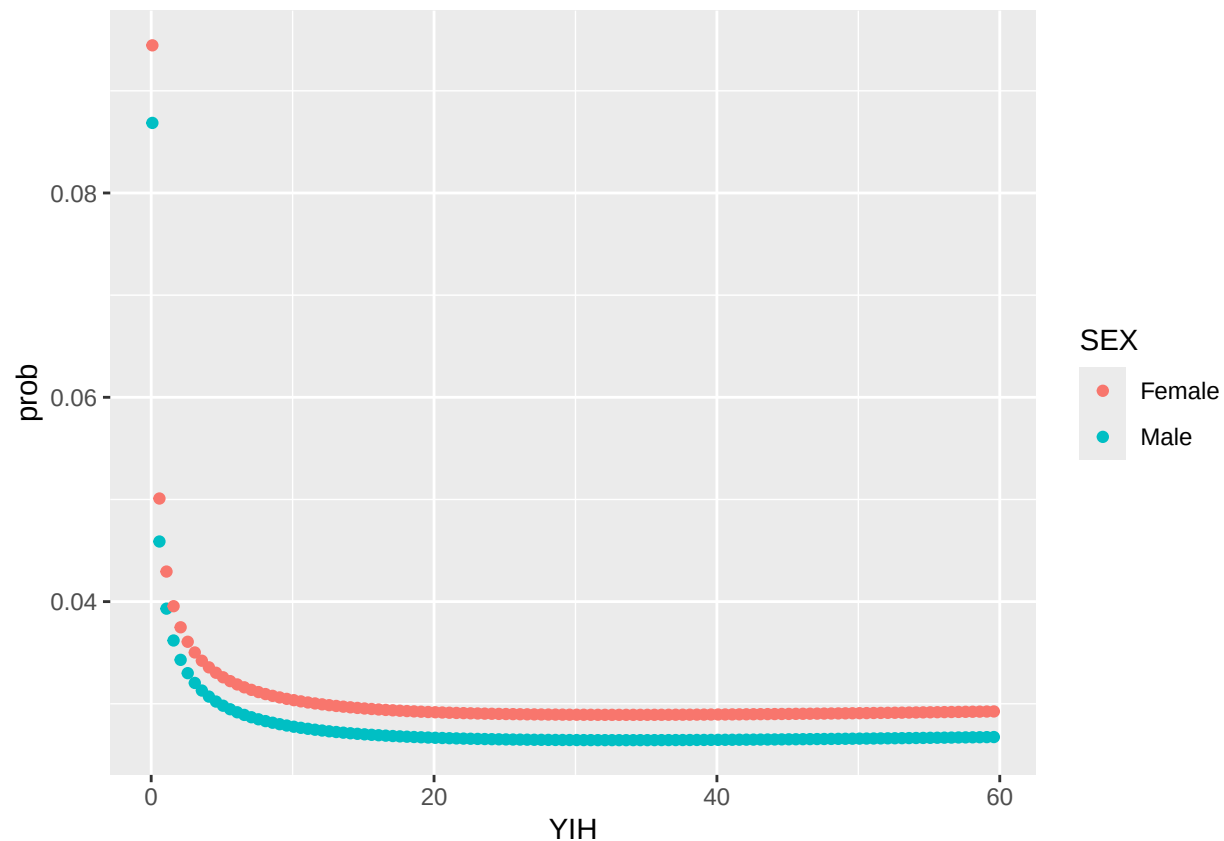
```
##           1           2           3           4           5           6
## 0.02703490 0.02727916 0.03291636 0.03176214 0.03364915 0.02702775
```

```
### Now we can plot the distributions for males v. females side by side.
### Note that, because females are predicted to report victimization
### more frequently, their distribution has shifted slightly to the right.
ggplot(data.frame("p" = c(m_pp, fm_pp),
                    "sex" = c(rep("Male", length(m_pp)),
                              rep("Female", length(fm_pp)))),
        aes(x = p, fill = sex)) +
  geom_density(alpha = 0.5)
```



```
## Calculating predicted probabilities for a range of values:
pp <- data.frame("YIH" = seq(min(person$YIH), max(person$YIH), by = 0.5) %>%
                  rep(each = 2),
                  "AGE" = mean(as.numeric(person$AGE)),
                  "EDUC" = mean(as.numeric(person$EDUC)),
                  "SEX" = as.factor(c("Male", "Female")))
pp$prob <- predict(m2, newdata = pp, type = "response")

ggplot(pp, aes(x = YIH, y = prob, color = SEX)) +
  geom_point()
```



```
### Note that this particular approach to visualizing your
### model predictions is particularly helpful for visualizing
### non-linear model parameters (and interactions).
```

```
### In general, generating model predictions are a very good
### way of understanding what a complex model might be telling
### you about your sample!
```

```
# ===== #
```

```
# Post-estimation functions
```

```
## Most of the same post-estimation functions that we used for OLS
## also apply to Logit (you should see some of them in the above code!)
## As a reminder:
```

```
### Akaike's Information Criterion (AIC)
```

```
AIC(m2)
```

```
## [1] 12682.24
```

```
### Bayesian Information Criterion (BIC)
```

```
BIC(m2)
```

```
## [1] 12734.07
```

```
### You can extract the coefficients as a named numeric vector:
coef(m2)
```

```
##           (Intercept)           log(YIH)           I(log(YIH)^2)
##           -3.17673709          -0.24488333           0.03491967
##  scale(as.numeric(AGE)) scale(as.numeric(EDUC))           SExFemale
##           -0.13625108           0.04197488           0.09214293
```

```
### You can generate a named matrix of confidence intervals:
confint(m2, level = 0.95)
```

```
## Waiting for profiling to be done...
```

```
##           2.5 %           97.5 %
## (Intercept)      -3.27671001 -3.07860829
## log(YIH)         -0.28609951 -0.20245085
## I(log(YIH)^2)     0.01498409  0.05448637
## scale(as.numeric(AGE)) -0.19512614 -0.07740210
## scale(as.numeric(EDUC)) -0.01337526  0.09723486
## SExFemale        -0.01194312  0.19654623
```

```
### You can generated a named numeric vector of predicted marginal scores:
fitted(m2) %>% head()
```

```
##           1           2           3           4           5           6
## 0.02703490 0.02983363 0.03291636 0.03176214 0.03677757 0.02702775
```

```
predict(m2) %>% head()
```

```
##           1           2           3           4           5           6
## -3.583220 -3.481831 -3.380315 -3.417203 -3.265396 -3.583491
```

```
### Analysis of Deviance:
anova(m2)
```

```
## Analysis of Deviance Table
```

```
##
```

```
## Model: binomial, link: logit
```

```
##
```

```
## Response: VIC
```

```
##
```

```
## Terms added sequentially (first to last)
```

```
##
```

```
##
```

```
##           Df Deviance Resid. Df Resid. Dev
## NULL                        41741      12851
## log(YIH)                   1  150.711      41740      12700
## I(log(YIH)^2)               1   6.817      41739      12694
## scale(as.numeric(AGE))      1  18.035      41738      12676
## scale(as.numeric(EDUC))     1   2.305      41737      12673
## SEX                         1   3.010      41736      12670
```

```
#### You can use this to compare model fit (Likelihood Ratio Test):
anova(m1, m2)
```

```
## Analysis of Variance Table
##
## Model 1: VIC ~ log(YIH) + I(log(YIH)^2) + scale(as.numeric(AGE)) + scale(as.numeric(EDUC)) +
##     SEX
## Model 2: VIC ~ log(YIH) + I(log(YIH)^2) + scale(as.numeric(AGE)) + scale(as.numeric(EDUC)) +
##     SEX
##   Res.Df    RSS Df Sum of Sq F Pr(>F)
## 1   41736  1428.5
## 2   41736 12670.2   0    -11242
```

```
#### You can print your variance-covariance matrix:
vcov(m2)
```

```
##               (Intercept)      log(YIH) I(log(YIH)^2)
## (Intercept)      0.0025535186 -1.040693e-04 -2.384825e-04
## log(YIH)         -0.0001040693  4.549539e-04 -1.136054e-04
## I(log(YIH)^2)    -0.0002384825 -1.136054e-04  1.014978e-04
## scale(as.numeric(AGE))  0.0004702201 -8.699884e-05 -6.141607e-05
## scale(as.numeric(EDUC)) -0.0001099118 -1.334927e-06  2.497381e-05
## SEXFemale        -0.0015376571  3.380997e-06 -4.622765e-06
##               scale(as.numeric(AGE)) scale(as.numeric(EDUC))
## (Intercept)           4.702201e-04      -1.099118e-04
## log(YIH)              -8.699884e-05      -1.334927e-06
## I(log(YIH)^2)         -6.141607e-05      2.497381e-05
## scale(as.numeric(AGE))  9.017117e-04      -2.977837e-04
## scale(as.numeric(EDUC)) -2.977837e-04      7.961737e-04
## SEXFemale            -2.488157e-05      -2.618781e-05
##               SEXFemale
## (Intercept)        -1.537657e-03
## log(YIH)           3.380997e-06
## I(log(YIH)^2)      -4.622765e-06
## scale(as.numeric(AGE)) -2.488157e-05
## scale(as.numeric(EDUC)) -2.618781e-05
## SEXFemale          2.827707e-03
```

```
#### The 'car' package will let you find the variance inflation factor (VIF):
vif(m2)
```

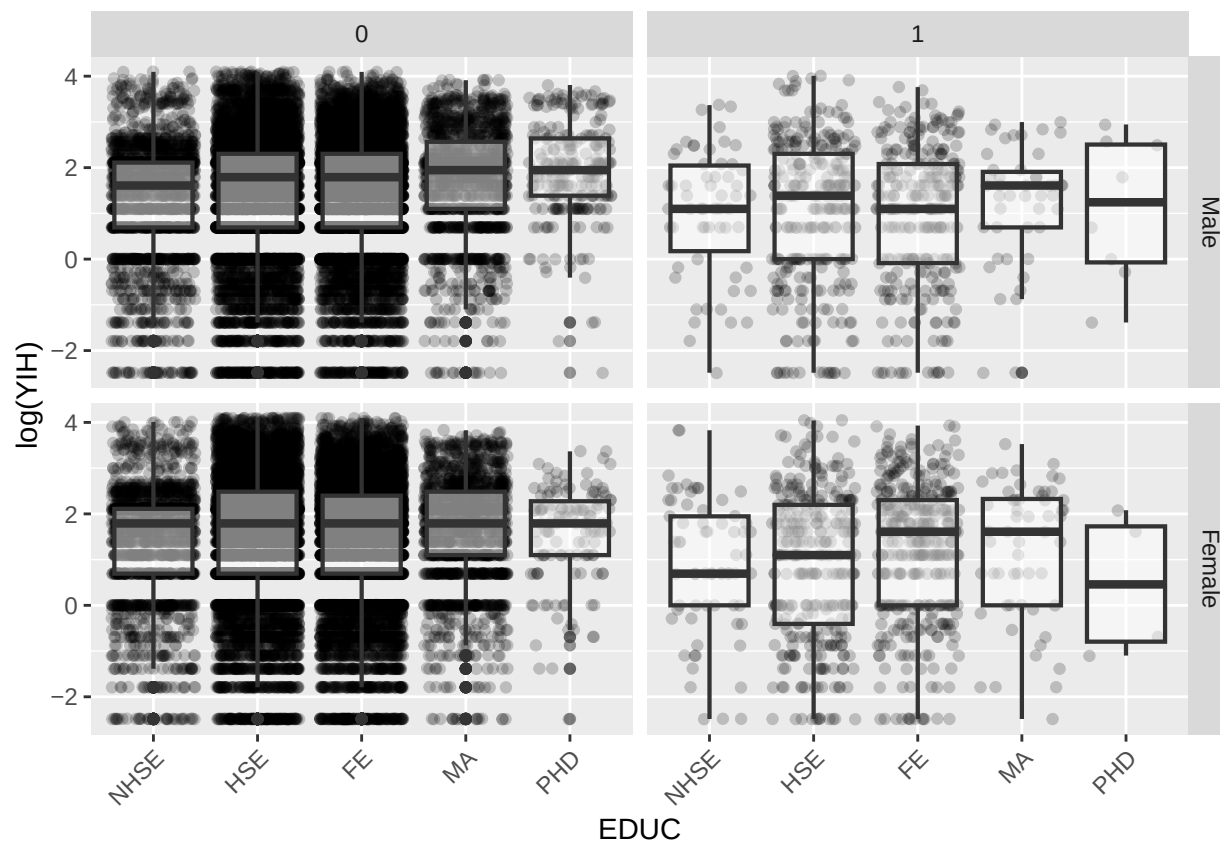
```
##           log(YIH)           I(log(YIH)^2)  scale(as.numeric(AGE))
##           1.522236           1.554570      1.292735
## scale(as.numeric(EDUC))           SEX
##           1.145225           1.001054
```

5 Estimating cumulative logit models

```
# Cumulative logit models can be understood as models that try to predict the
# distribution of complex crosstabs. If, for example, we wanted to examine
# the effect of victimization (1 = experienced victimization), and sex
# (2 = Female; 1 = Male). We would start with a crosstab of all 3 variables:
fable(xtabs(~ EDUC + VIC + SEX, data = person))
```

```
##          SEX Male Female
## EDUC VIC
## NHSE 0      1997    1763
##       1         58     66
## HSE  0      7141    8079
##       1        283    315
## FE   0      8201    9593
##       1        279    374
## MA   0      1470    1645
##       1         46     56
## PHD  0        238    126
##       1          8     4
```

```
# For obvious reasons, this only works with binary independent variables
# (or categorical variables with relatively few categories). If we have a
# continuous variable we wanted to introduce into the mix, we could instead
# plot the distribution across all four variables with box plots:
ggplot(person, aes(x = EDUC, y = log(YIH))) +
  geom_jitter(alpha = 0.2) +
  geom_boxplot(size = 0.75, alpha = 0.5) +
  facet_grid(SEX ~ VIC) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, vjust = 1))
```



```
# To run a ordered / cumulative / ordinal logit, you need to import
# the MASS package:
library(MASS)
```

```
##
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
##
## select
```

```
# Otherwise, it follows much the same syntax as the OLS and Logit models:
summary(m3 <- polr(EDUC ~ VIC + SEX + YIH, data = person, Hess = TRUE))
```

```
## Call:
## polr(formula = EDUC ~ VIC + SEX + YIH, data = person, Hess = TRUE)
##
## Coefficients:
##              Value Std. Error t value
## VIC          -0.0393429  0.0489408  -0.8039
## SEXFemale    0.0686523  0.0183576   3.7397
## YIH          0.0006004  0.0009676   0.6205
##
## Intercepts:
```

```
##           Value      Std. Error t value
## NHSE|HSE   -2.2373      0.0210  -106.4022
## HSE|FE     -0.0718      0.0161   -4.4724
## FE|MA       2.4032      0.0216   111.0563
## MA|PHD      4.7413      0.0534    88.8429
##
## Residual Deviance: 99289.60
## AIC: 99303.60
```

Unlike these other models, the polr() function does not calculate p-values for you (remember: p-values are not 'native' to the 'regression' process whereby a line/plane of best fit is fit to probability distributions).

You will need to use this code to calculate your own:

Extract model coefficients:

```
(ctable <- coef(summary(m3)))
```

```
##           Value      Std. Error      t value
## VIC        -0.0393429500 0.0489407957  -0.8038886
## SEXFemale   0.0686523013 0.0183575975   3.7397215
## YIH         0.0006004283 0.0009675851   0.6205431
## NHSE|HSE    -2.2373031927 0.0210268442 -106.4022337
## HSE|FE      -0.0718352292 0.0160618480   -4.4724137
## FE|MA       2.4031660229 0.0216391622   111.0563340
## MA|PHD      4.7412540346 0.0533667321    88.8428773
```

Calculate and store p values:

```
p <- pnorm(abs(ctable[, "t value"]), lower.tail = FALSE) * 2
```

Generate significance stars:

```
s <- ifelse(p <= 0.001, "***",
            ifelse(p <= 0.01, "**",
                    ifelse(p <= 0.05, "*", "n.s.")))
```

Add p-values to the model coefficient data frame:

```
(ctable <- cbind(round(ctable, 3), `Pr(>|t|)` = round(p, 3), `sig` = s))
```

```
##           Value      Std. Error t value      Pr(>|t|) sig
## VIC        "-0.039" "0.049"      "-0.804" "0.421" "n.s."
## SEXFemale   "0.069" "0.018"      "3.74"    "0"     "***"
## YIH         "0.001" "0.001"      "0.621"  "0.535" "n.s."
## NHSE|HSE    "-2.237" "0.021"     "-106.402" "0"     "***"
## HSE|FE      "-0.072" "0.016"     "-4.472"  "0"     "***"
## FE|MA       "2.403" "0.022"     "111.056" "0"     "***"
## MA|PHD      "4.741" "0.053"     "88.843"  "0"     "***"
```

We can also extract the confidence intervals how we would a logit:

```
confint(m3)
```

```
## Waiting for profiling to be done...
```

```
##           2.5 %      97.5 %
## VIC        -0.135252665 0.056611562
## SEXFemale   0.032675438 0.104632708
## YIH        -0.001293111 0.002498701
```

```
# Then, if we want to present our results as odds ratios, we can apply
# the following code to generate the OR and the 95% CI:
```

```
exp(cbind(OR = coef(m3), confint(m3)))
```

```
## Waiting for profiling to be done...
```

```
##              OR      2.5 %   97.5 %
## VIC      0.9614209 0.8734952 1.058245
## SEXFemale 1.0710637 1.0332151 1.110303
## YIH      1.0006006 0.9987077 1.002502
```

```
# Testing the proportional odds assumption
```

```
## The following code generates the log-linear parameter estimates
## for each independent variable as if we had regressed each level
## of the dependent on each independent variable individually:
```

```
sf <- function(y) {
  c('Y>=1' = qlogis(mean(y >= 1)),
    'Y>=2' = qlogis(mean(y >= 2)),
    'Y>=3' = qlogis(mean(y >= 3)),
    'Y>=4' = qlogis(mean(y >= 4)),
    'Y>=5' = qlogis(mean(y >= 5)))
}
```

```
(s <- with(person, summary(as.numeric(EDUC) ~ VIC + SEX + log(YIH), fun = sf)))
```

```
## as.numeric(EDUC)      N= 41742
```

```
##
## +-----+-----+-----+-----+-----+-----+-----+
## |          |          | N|Y>=1|   Y>=2|   Y>=3|   Y>=4|   Y>=5|
## +-----+-----+-----+-----+-----+-----+-----+
## |    VIC|      No|40253| Inf|2.272702|0.114052870|-2.358046|-4.696702|
## |          |      Yes| 1489| Inf|2.398628|0.060461662|-2.490011|-4.812862|
## +-----+-----+-----+-----+-----+-----+-----+
## |    SEX|      Male|19721| Inf|2.151366|0.077418088|-2.321642|-4.371555|
## |          |     Female|22021| Inf|2.401517|0.143289942|-2.400325|-5.126296|
## +-----+-----+-----+-----+-----+-----+-----+
## |log(YIH)|[-2.48,1.10]|12163| Inf|2.227086|0.005426306|-2.625745|-5.273909|
## |          |[ 1.10,1.95]|10433| Inf|2.248646|0.244477003|-2.299636|-4.453669|
## |          |[ 1.95,2.48]| 9693| Inf|2.107960|0.080720579|-2.324224|-4.670021|
## |          |[ 2.48,4.09]| 9453| Inf|2.599286|0.136676985|-2.183547|-4.498634|
## +-----+-----+-----+-----+-----+-----+-----+
## | Overall|          |41742| Inf|2.276977|0.112138831|-2.362512|-4.700625|
## +-----+-----+-----+-----+-----+-----+-----+
```

```
## Compare with the results of the following logit models:
```

```
glm(I(as.numeric(EDUC) >= 2) ~ VIC, family = "binomial", data = person)
```

```
##
## Call:  glm(formula = I(as.numeric(EDUC) >= 2) ~ VIC, family = "binomial",
##        data = person)
##
```

```
## Coefficients:
## (Intercept)          VIC
##      2.2727      0.1259
##
## Degrees of Freedom: 41741 Total (i.e. Null);  41740 Residual
## Null Deviance:      25840
## Residual Deviance: 25840      AIC: 25840
```

```
glm(I(as.numeric(EDUC) >= 3) ~ VIC, family = "binomial", data = person)
```

```
##
## Call: glm(formula = I(as.numeric(EDUC) >= 3) ~ VIC, family = "binomial",
##      data = person)
##
## Coefficients:
## (Intercept)          VIC
##      0.11405      -0.05359
##
## Degrees of Freedom: 41741 Total (i.e. Null);  41740 Residual
## Null Deviance:      57740
## Residual Deviance: 57730      AIC: 57740
```

```
glm(I(as.numeric(EDUC) >= 4) ~ VIC, family = "binomial", data = person)
```

```
##
## Call: glm(formula = I(as.numeric(EDUC) >= 4) ~ VIC, family = "binomial",
##      data = person)
##
## Coefficients:
## (Intercept)          VIC
##      -2.358      -0.132
##
## Degrees of Freedom: 41741 Total (i.e. Null);  41740 Residual
## Null Deviance:      24490
## Residual Deviance: 24490      AIC: 24490
```

```
glm(I(as.numeric(EDUC) >= 5) ~ VIC, family = "binomial", data = person)
```

```
##
## Call: glm(formula = I(as.numeric(EDUC) >= 5) ~ VIC, family = "binomial",
##      data = person)
##
## Coefficients:
## (Intercept)          VIC
##      -4.6967      -0.1162
##
## Degrees of Freedom: 41741 Total (i.e. Null);  41740 Residual
## Null Deviance:      4290
## Residual Deviance: 4290      AIC: 4294
```

Now, let's see if each "step up" through the dependent variable (for
each independent variable) is approximately equidistant. Ideally,
we want the values here to all be approximately equal.

```
s[, 4] - s[, 3]
```

```
## as.numeric(EDUC)      N= 41742
##
## +-----+-----+-----+
## |          |          | Y>=3 |
## +-----+-----+-----+
## |    VIC   |    No  |-2.158649|
## |          |    Yes |-2.338166|
## +-----+-----+-----+
## |    SEX   |    Male|-2.073948|
## |          |    Female|-2.258227|
## +-----+-----+-----+
## |log(YIH) | [-2.48,1.10] |-2.221660|
## |          | [ 1.10,1.95] |-2.004169|
## |          | [ 1.95,2.48] |-2.027239|
## |          | [ 2.48,4.09] |-2.462609|
## +-----+-----+-----+
## | Overall |          |-2.164838|
## +-----+-----+-----+
```

```
s[, 5] - s[, 4]
```

```
## as.numeric(EDUC)      N= 41742
##
## +-----+-----+-----+
## |          |          | Y>=4 |
## +-----+-----+-----+
## |    VIC   |    No  |-2.472099|
## |          |    Yes |-2.550472|
## +-----+-----+-----+
## |    SEX   |    Male|-2.399060|
## |          |    Female|-2.543615|
## +-----+-----+-----+
## |log(YIH) | [-2.48,1.10] |-2.631171|
## |          | [ 1.10,1.95] |-2.544113|
## |          | [ 1.95,2.48] |-2.404945|
## |          | [ 2.48,4.09] |-2.320224|
## +-----+-----+-----+
## | Overall |          |-2.474651|
## +-----+-----+-----+
```

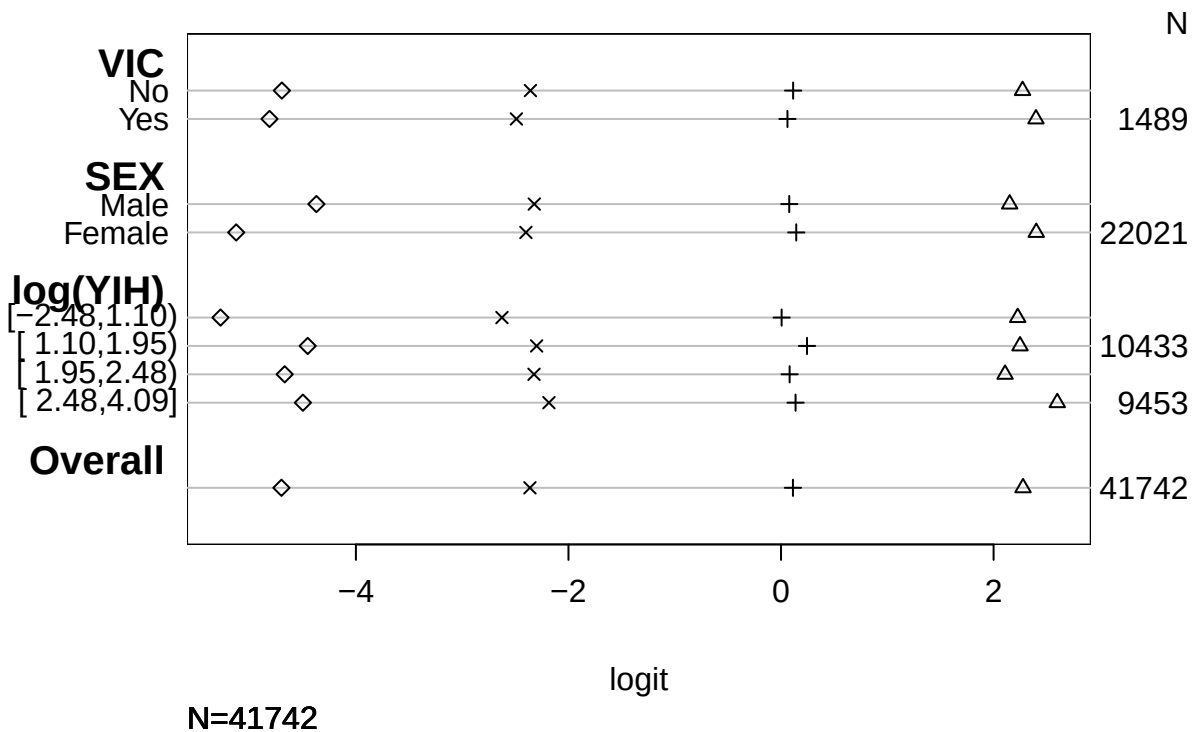
```
s[, 6] - s[, 5]
```

```
## as.numeric(EDUC)      N= 41742
##
## +-----+-----+-----+
## |          |          | Y>=5 |
## +-----+-----+-----+
```

```
## |      VIC|      No|-2.338656|
## |      |      Yes|-2.322851|
## +-----+-----+-----+
## |      SEX|      Male|-2.049913|
## |      |      Female|-2.725971|
## +-----+-----+-----+
## |log(YIH)|[-2.48,1.10)|-2.648164|
## |      | [ 1.10,1.95)|-2.154032|
## |      | [ 1.95,2.48)|-2.345797|
## |      | [ 2.48,4.09]|-2.315086|
## +-----+-----+-----+
## | Overall|      |-2.338113|
## +-----+-----+-----+
```

For convenience, it is easier to generate a plot that visualizes the results:

```
plot(s,
      which = 1:5,
      pch = 1:5,
      xlab = 'logit',
      main = ' ',
      xlim = range(s[, 3:6]))
```



However, for peer-reviewed publication, I would recommend performing the
Brant-Wald Test to make sure that the proportional odds assumption holds:
library(brant)

```
## Warning: package 'brant' was built under R version 4.3.3
```

```
brant(m3)
```

```
## -----  
## Test for X2 df probability  
## -----  
## Omnibus      233.53  9    0  
## VIC          5.37   3   0.15  
## SEXFemale    95.15  3    0  
## YIH          133.98 3    0  
## -----  
##  
## H0: Parallel Regression Assumption holds
```

```
### A low p-value for a given independent variable implies that, for that  
### variable, the proportional odds assumption does not hold. So, according  
### to the results, only the "Victimization" variable satisfies the assumption.  
### This COULD imply that the cumulative logit model is a bad 'fit' for the  
### "Education Level" dependent variable. However, if you look at our previous  
### figure, the points on the line are evenly dispersed along the x-axis.
```

```
### One important thing to remember about p-values is that they rely on the  
### standard error, which is 'penalized' by the sample size. For particularly  
### large samples - like this one (N = 41,742) - the test might be overpowered  
### and thus able to identify very small deviations that would, for smaller  
### sample sizes, not "flag" as significant (this ALSO applies to the p-values  
### of regression coefficients!)
```

```
### If you are worried about your sample size, you could perform the same  
### analysis on a subset of your data to check the "sensitivity" of the test  
### to your sample size. Here we are going to perform the same analysis on a  
### random set of observations (5%; N = 2,077):
```

```
sub_person <- person[sample(nrow(person), size = floor(nrow(person) * 0.05)), ]
```

```
summary(m4 <- polr(EDUC ~ VIC + SEX + YIH,  
                  data = sub_person,  
                  Hess = TRUE))
```

```
## Call:
```

```
## polr(formula = EDUC ~ VIC + SEX + YIH, data = sub_person, Hess = TRUE)
```

```
##
```

```
## Coefficients:
```

```
##           Value Std. Error t value  
## VIC      -0.164135  0.217897 -0.7533  
## SEXFemale -0.149181  0.082111 -1.8168  
## YIH       -0.001335  0.004446 -0.3003
```

```
##
```

```
## Intercepts:
```

```
##           Value Std. Error t value  
## NHSE|HSE -2.3470  0.0947 -24.7778  
## HSE|FE    -0.1778  0.0716  -2.4823  
## FE|MA      2.3707  0.0978  24.2298
```

```
## MA|PHD      4.9071    0.2736    17.9340
##
## Residual Deviance: 4901.546
## AIC: 4915.546
```

```
brant(m4)
```

```
## -----
## Test for X2  df  probability
## -----
## Omnibus      19.99   9   0.02
## VIC          1.55    3   0.67
## SEXFemale     5.01    3   0.17
## YIH          13.72    3    0
## -----
##
## H0: Parallel Regression Assumption holds
```

As you can see, the significance test maintains that the parallel odds assumption does NOT hold. This implies that the relationships between "Sex" and "Years in Household" are likely NOT the same for each response category of the dependent variable (Education Level).

In other words, we need "random slopes" by group (this could be achieved using multi-level or "random effects" models that are introduced later in this course!)

6 Estimating multinomial logit models

```
# Load the nnet package (which includes multinomial logit models!)
library(nnet)
```

```
## Warning: package 'nnet' was built under R version 4.3.3
```

```
# Prepare the data:
## 1. Filter to the victimized population.
## 2. Select the variables you plan on using.
## 3. Assign to a new object, mn_pers.
mn_pers <- person %>%
  filter(!is.na(VIC_LOC)) %>%
  dplyr::select(VIC_LOC, VIOLENT, AGE, SEX, EDUC, YIH)
```

```
# Prepare the dependent variable (optional):
## Manually set the levels of the variable
## so that "other" is the reference category.
mn_pers$VIC_LOC <- factor(mn_pers$VIC_LOC, levels = c("Home",
  "Communal Area",
  "Open Area",
  "Other"))
```

```

# Fit the multinomial logit model:
summary(m5 <- multinom(VIC_LOC ~ VIOLENT +
                        as.numeric(AGE) +
                        SEX +
                        as.numeric(EDUC) +
                        YIH, data = mn_pers))

## # weights: 28 (18 variable)
## initial value 2064.192304
## iter 10 value 1632.520064
## iter 20 value 1591.449997
## final value 1588.680718
## converged

## Call:
## multinom(formula = VIC_LOC ~ VIOLENT + as.numeric(AGE) + SEX +
##          as.numeric(EDUC) + YIH, data = mn_pers)
##
## Coefficients:
##          (Intercept)  VIOLENT as.numeric(AGE)  SEXFemale
## Communal Area  0.003443909 0.2917761      -0.34562234 -0.01882452
## Open Area     -1.304181850 0.2043588      -0.08596514 -0.36274607
## Other         -2.162478322 0.1676364      -0.07402919 -0.31191672
##          as.numeric(EDUC)          YIH
## Communal Area      0.06331532  0.011331015
## Open Area          0.35263866 -0.023804505
## Other              0.17444565  0.006477163
##
## Std. Errors:
##          (Intercept)  VIOLENT as.numeric(AGE) SEXFemale as.numeric(EDUC)
## Communal Area  0.2573169 0.09576168      0.04063403 0.1489641      0.10712188
## Open Area      0.2705638 0.10083771      0.03781041 0.1397479      0.09690347
## Other          0.4116665 0.14832936      0.05759330 0.2152484      0.14992421
##          YIH
## Communal Area 0.009816807
## Open Area     0.010301925
## Other         0.012567325
##
## Residual Deviance: 3177.361
## AIC: 3213.361

# Output the results to a format that is easily introduced
# into a manuscript document or presentation:
## Load the broom and kableExtra libraries
library(broom)

```

```
## Warning: package 'broom' was built under R version 4.3.3
```

```
library(kableExtra)
```

```
## Warning: package 'kableExtra' was built under R version 4.3.3
```

```
##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```
## Convert the model results into a data frame format:
tidy(m5, conf.int = TRUE)
```

```
## # A tibble: 18 x 8
##   y.level      term estimate std.error statistic  p.value conf.low conf.high
##   <chr>        <chr>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 Communal Area (Inte~  0.00344  0.257      0.0134 9.89e- 1 -0.501    0.508
## 2 Communal Area VIOLE~  0.292    0.0958     3.05  2.31e- 3  0.104    0.479
## 3 Communal Area as.nu~ -0.346    0.0406    -8.51  1.80e-17 -0.425   -0.266
## 4 Communal Area SEXFe~ -0.0188   0.149     -0.126 8.99e- 1 -0.311    0.273
## 5 Communal Area as.nu~  0.0633   0.107      0.591 5.54e- 1 -0.147    0.273
## 6 Communal Area YIH     0.0113   0.00982    1.15  2.48e- 1 -0.00791  0.0306
## 7 Open Area      (Inte~ -1.30     0.271     -4.82  1.43e- 6 -1.83     -0.774
## 8 Open Area      VIOLE~  0.204     0.101      2.03  4.27e- 2  0.00672  0.402
## 9 Open Area      as.nu~ -0.0860   0.0378    -2.27  2.30e- 2 -0.160   -0.0119
## 10 Open Area     SEXFe~ -0.363    0.140     -2.60  9.44e- 3 -0.637   -0.0888
## 11 Open Area     as.nu~  0.353    0.0969     3.64  2.74e- 4  0.163    0.543
## 12 Open Area     YIH     -0.0238   0.0103    -2.31  2.09e- 2 -0.0440  -0.00361
## 13 Other         (Inte~ -2.16     0.412     -5.25  1.50e- 7 -2.97     -1.36
## 14 Other         VIOLE~  0.168     0.148      1.13  2.58e- 1 -0.123    0.458
## 15 Other         as.nu~ -0.0740   0.0576    -1.29  1.99e- 1 -0.187    0.0389
## 16 Other         SEXFe~ -0.312    0.215     -1.45  1.47e- 1 -0.734    0.110
## 17 Other         as.nu~  0.174     0.150      1.16  2.45e- 1 -0.119    0.468
## 18 Other         YIH     0.00648   0.0126     0.515 6.06e- 1 -0.0182  0.0311
```

```
## Data frame format with relative risk ratios:
tidy(m5, conf.int = TRUE, exponentiate = TRUE)
```

```
## # A tibble: 18 x 8
##   y.level      term estimate std.error statistic  p.value conf.low conf.high
##   <chr>        <chr>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 Communal Area (Inte~  1.00     0.257      0.0134 9.89e- 1  0.606    1.66
## 2 Communal Area VIOLE~  1.34     0.0958     3.05  2.31e- 3  1.11     1.62
## 3 Communal Area as.nu~  0.708    0.0406    -8.51  1.80e-17  0.654    0.766
## 4 Communal Area SEXFe~  0.981    0.149     -0.126 8.99e- 1  0.733    1.31
## 5 Communal Area as.nu~  1.07     0.107      0.591 5.54e- 1  0.864    1.31
## 6 Communal Area YIH     1.01     0.00982    1.15  2.48e- 1  0.992    1.03
## 7 Open Area      (Inte~  0.271    0.271     -4.82  1.43e- 6  0.160    0.461
## 8 Open Area      VIOLE~  1.23     0.101      2.03  4.27e- 2  1.01     1.49
## 9 Open Area      as.nu~  0.918    0.0378    -2.27  2.30e- 2  0.852    0.988
## 10 Open Area     SEXFe~  0.696    0.140     -2.60  9.44e- 3  0.529    0.915
## 11 Open Area     as.nu~  1.42     0.0969     3.64  2.74e- 4  1.18     1.72
## 12 Open Area     YIH     0.976    0.0103    -2.31  2.09e- 2  0.957    0.996
## 13 Other         (Inte~  0.115    0.412     -5.25  1.50e- 7  0.0513   0.258
## 14 Other         VIOLE~  1.18     0.148      1.13  2.58e- 1  0.884    1.58
```

```
## 15 Other      as.nu~    0.929   0.0576   -1.29   1.99e- 1   0.830     1.04
## 16 Other      SEXFe~    0.732   0.215    -1.45   1.47e- 1   0.480     1.12
## 17 Other      as.nu~    1.19    0.150     1.16   2.45e- 1   0.887     1.60
## 18 Other      YIH       1.01    0.0126    0.515   6.06e- 1   0.982     1.03
```

```
## Output the data frame to a kable (html table):
tidy(m5, conf.int = TRUE, exponentiate = TRUE) %>%
  kable() %>%
  kable_styling("basic", full_width = FALSE)
```

y.level	term	estimate	std.error	statistic	p.value	conf.low	conf.high
Communal Area	(Intercept)	1.0034498	0.2573169	0.0133839	0.9893215	0.6059923	1.6615915
Communal Area	VIOLENT	1.3388033	0.0957617	3.0468984	0.0023122	1.1096966	1.6152110
Communal Area	as.numeric(AGE)	0.7077797	0.0406340	-8.5057368	0.0000000	0.6535975	0.7664536
Communal Area	SEXFemale	0.9813516	0.1489641	-0.1263695	0.8994394	0.7328687	1.3140838
Communal Area	as.numeric(EDUC)	1.0653627	0.1071219	0.5910588	0.5544810	0.8636051	1.3142555
Communal Area	YIH	1.0113955	0.0098168	1.1542465	0.2483991	0.9921216	1.0310437
Open Area	(Intercept)	0.2713945	0.2705638	-4.8202373	0.0000014	0.1596970	0.4612171
Open Area	VIOLENT	1.2267382	0.1008377	2.0266107	0.0427022	1.0067431	1.4948070
Open Area	as.numeric(AGE)	0.9176262	0.0378104	-2.2735840	0.0229910	0.8520823	0.9882119
Open Area	SEXFemale	0.6957631	0.1397479	-2.5957169	0.0094394	0.5290634	0.9149872
Open Area	as.numeric(EDUC)	1.4228169	0.0969035	3.6390717	0.0002736	1.1766970	1.7204157
Open Area	YIH	0.9764766	0.0103019	-2.3106852	0.0208503	0.9569579	0.9963934
Other	(Intercept)	0.1150397	0.4116665	-5.2529859	0.0000001	0.0513377	0.2577857
Other	VIOLENT	1.1825066	0.1483294	1.1301633	0.2584074	0.8841896	1.5814728
Other	as.numeric(AGE)	0.9286446	0.0575933	-1.2853787	0.1986600	0.8295184	1.0396162
Other	SEXFemale	0.7320425	0.2152484	-1.4491014	0.1473093	0.4800832	1.1162360
Other	as.numeric(EDUC)	1.1905860	0.1499242	1.1635589	0.2446028	0.8874524	1.5972633
Other	YIH	1.0064982	0.0125673	0.5153971	0.6062756	0.9820095	1.0315976